

Depot

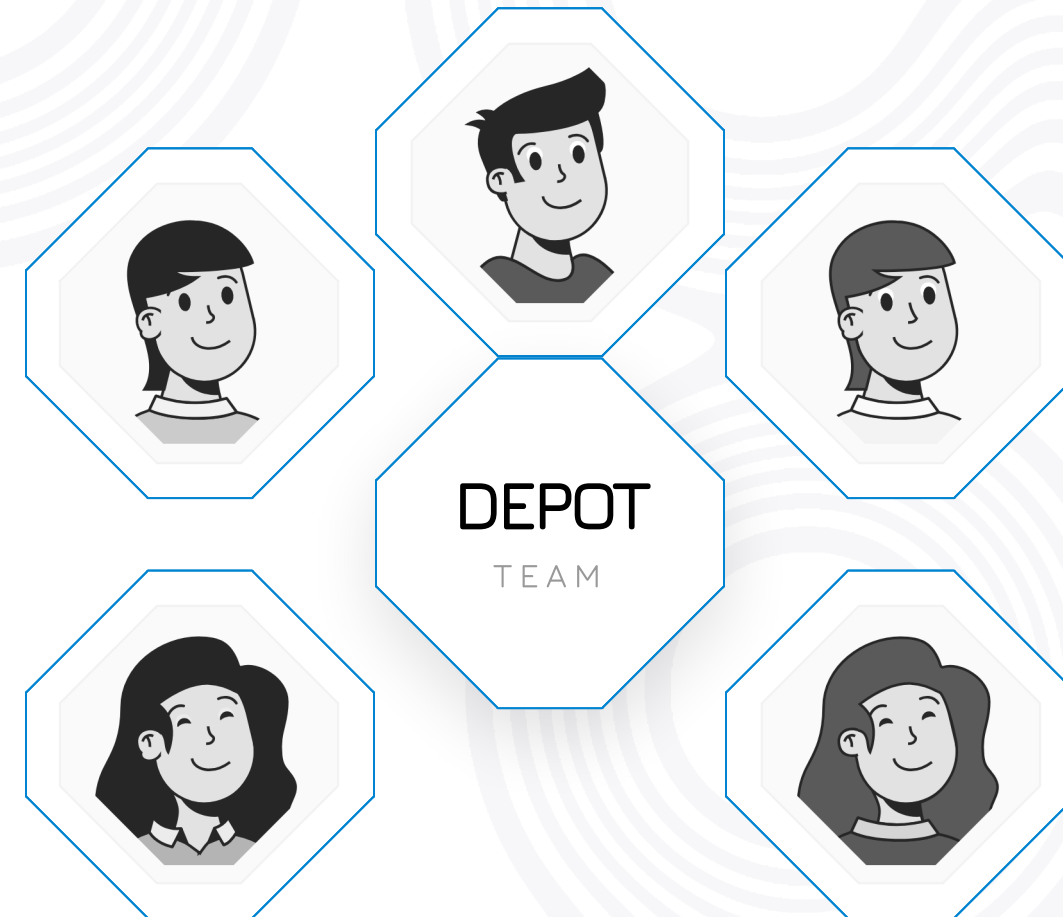
An AI-powered infrastructure architect,
from code to cloud.

TEAM

Ahsan Riaz
Muhammad Waleed
Muhammad Faizan Anwar

ADVISORS

Hira Anwar
Ayesha Hakim



Building is fast now. The **infrastructure** is still expert work.

~95%

of the code was AI-generated in a quarter of YC's Winter 2025 start-ups [1]

150%

average yearly growth in infrastructure-as-code questions on Stack Overflow, 2011–2022 [2]

0%

of one model's Terraform passed a security scan, though 77.8% passed validation [3]

Working code still leaves four decisions.

01

What it needs

02

Where it runs

03

What it costs

04

How to change
it safely

Depot answers all four, **and explains, records and approves every change.**

Each tool solves part of it.

Someone still decides where it runs.

	Vercel managed platform	Flightcontrol bring-your-own-cloud	Encore framework-driven	Pulumi Neo AI and enterprise IaC	Depot proposed
Starts from code as it is	Plus platform config	Repo plus config file	Encore primitives	Pulumi estate	Ordinary repository
Runs in the user's account	No — their platform	Yes	Yes	Yes	Yes
Chooses cloud or platform	The platform only	AWS only	AWS and GCP	Fixed set, org-led	Compares all five
Approval before change	No approval gate	Deploy only	Not stated	Platform team's call	Signed, risk-tiered

Built for builders without an infrastructure engineer.



AI-assisted builders

A working product, but little experience of running one.



Solo full-stack developers

Outgrowing a managed platform: stay, or move to a cloud?



Small teams and agencies

Two to ten people, with no DevOps staff.

Three objectives.

What a user can do with Depot.

01

Deploy any supported repository

Point Depot at GitHub, GitLab or an archive. It reads the application, decides the target and pattern, prices it, then provisions it to AWS, GCP, Azure, Vercel or Railway inside the user's own account — every change validated, costed and signed before it runs.

02

Check health, metrics and cost

Deployment history, health checks, logs, metrics, environment-variable metadata and running spend in one place, with rollback and the savings found. The agent investigates an incident and proposes a fix; a person approves.

03

Review every change, and trace it after

Nothing reaches the cloud without a saved plan you can read and an explicit approval. Every release then links back through the commit, the model, the specification, the plan, the workflow run and the health outcome — a reviewer follows the whole chain, not just a diff.

Research sits inside all three: comparing language models on extraction, and measured guardrails for the agent loop.

What Depot actually does.



Application Model

Every fact marked detected, inferred or user-supplied.



No stored keys

Bounded roles, federation, or one scoped token per target.



Architect

Picks among reviewed patterns, ranked by cost and effort.



Delivery and operations

Deploy on push, rollback, logs, cost, alerts, savings.



Safety gate

Validate, saved plan, policy, cost and role check.



Agent and incident loop

Changes by conversation; alarms end in a fix a person approves.

One session, from working code to live.

STEP 1

Connect

GitHub, GitLab or an
archive

STEP 2

Answer

Up to eight
questions

STEP 3

Review

Target, architecture,
monthly cost

STEP 4

Approve

Signed and role-
checked

STEP 5

Run

Deploy on push, roll
back, watch cost

In scope, and deliberately out.

IN SCOPE

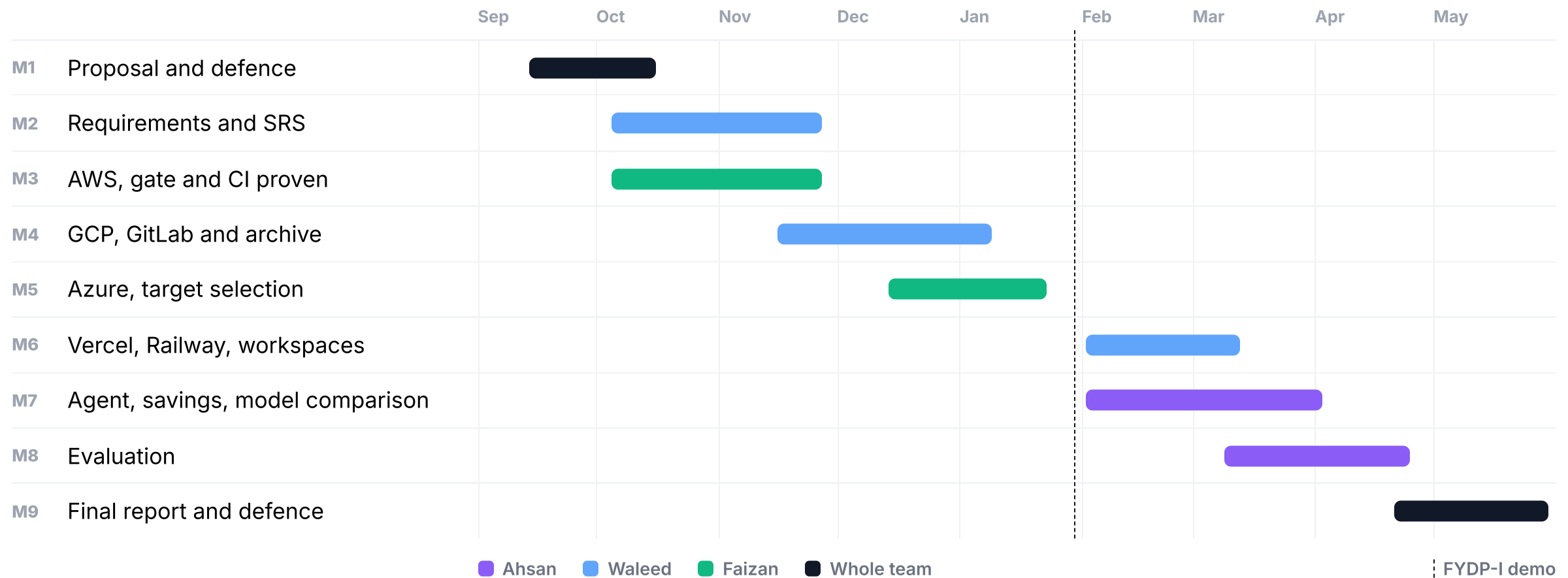
- ✓ GitHub, GitLab or an archive
- ✓ Web apps, APIs, static sites and workers (JS/TS, Python, Go)
- ✓ AWS, GCP, Azure, Vercel, Railway
- ✓ Scoped, short-lived credentials
- ✓ Approval before every infrastructure change
- ✓ Delivery, rollback, monitoring, cost, savings, agent

OUT OF SCOPE

- ✗ GPU, mobile and desktop software
- ✗ Clouds without a reviewed pattern library
- ✗ Pasted long-lived keys
- ✗ Any unapproved change
- ✗ Compliance certification

Extended scope only if M8 is met early: guided mode, Oracle and Yandex Cloud, voice, preview environments.

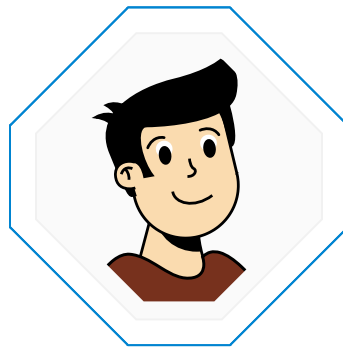
Three clouds first, then everything else.



Team planning targets; dates will follow the official defence schedule.

The Architects

Three owners. No member reviews only their own work.



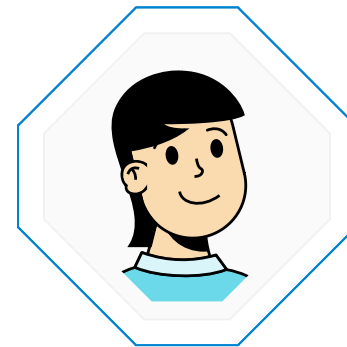
Ahsan Riaz

Application Model, agent mode,
incident loop, research

01

02

03

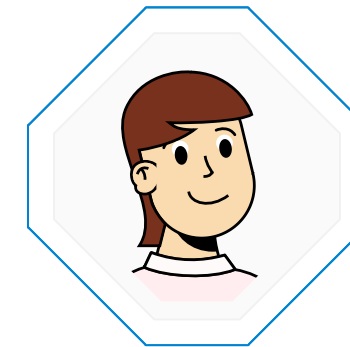


Muhammad Waleed

User-facing work, team workspaces,
managed platforms

01

02



Muhammad Faizan Anwar

Architect, cloud adapters, safety gate,
savings

01

03

Advisor **Hira Anwar** · Co-advisor **Ayesha Hakim** · Scrum, two-week sprints

What could go wrong, and how we handle it.

Cloud accounts or credits not ready

First-sprint task with an owner; no features until fixed

Scope outgrows three people

Extended scope only if M8 is met early; Railway can move out

AI proposes unsafe infrastructure

The model only selects; reviewed templates execute

An agent is tricked into acting

Fixed actions, fixed permissions, approval by risk [5]

Credentials reach too far

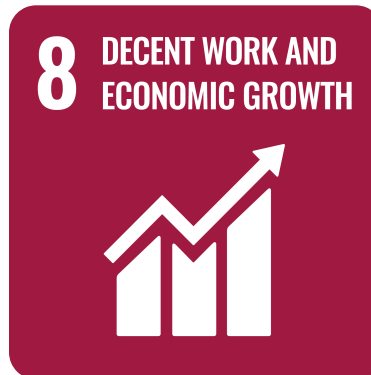
Bounded, short-lived roles from M3; no stored user keys

A member acts without the role

Server-side role check on every route

Every change is traceable to its model, plan and approval.

Two goals we contribute to.



SDG 8.3

Small-enterprise growth

Small teams run products on cloud accounts they own, at a known monthly cost, without hiring.



SDG 9.5

Research and technological capability

An open benchmark, and views that explain why each architecture was chosen.

Sources.

1

I. Mehta, "A quarter of startups in YC's current cohort have codebases that are almost entirely AI-generated," TechCrunch, Mar. 2025.

2

M. Begoug et al., "What do infrastructure-as-code practitioners discuss: An empirical study on Stack Overflow," ESEM, 2023.

3

F. L. S. Vargas, R. B. Mansilha and D. Kreutz, "Security-first evaluation of text-to-Terraform," arXiv:2608.02672, 2026.

4

Vendor documentation of the nine products compared, accessed Sep. 30, 2026.

5

OWASP Gen AI Security Project, "LLM06:2025 Excessive agency," 2025.

6

United Nations, "Goal 8" and "Goal 9," Sustainable Development Goals, accessed Sep. 30, 2026.

Thank you.

Any questions?

Ahsan Riaz

Muhammad Waleed

Muhammad Faizan Anwar

Advisor Hira Anwar · Co-advisor Ayesha Hakim · NUST SEECS